



IBM BOB 2.0 HACKATHON • TEAM NORTH

# Linty

## AI code review that **proves its fixes.**

Four IBM Bob agents find the bugs, fix them, write the tests, and then run those tests to prove it.

# Code review is slow, manual, and the tests get skipped.



## Reviews eat hours

A reviewer reads every line, spots the bug, explains it, and waits for a fix. It happens on every pull request.



## Tests get skipped

Writing regression tests is the first step cut under deadline pressure, so the same bug comes back.



## AI fixes aren't trusted

Assistants suggest fixes, but without proof a human still has to verify every change by hand.

# One command. Four agents. **A verified review.**



```
/linty-review samples/cart-total.ts
```

# Linty ships as a Bob configuration.

## Custom modes

An orchestrator plus four specialist modes. Each can only write its own output file (fileRegex): permissions as governance.

## Subtasks

The orchestrator hands each stage to a specialist as a separate Bob subtask, a genuine multi-agent pipeline.

## Slash command

`/linty-review <file>` runs everything: start, 4 agents, finalize, summary.

## Rules + .bobignore

Shared guardrails for every mode: no secrets, never touch the ground truth, report measured numbers only.

# Every fix ships with tests that were actually run.

3/12

tests pass on the **original** buggy code

12/12

tests pass on Linty's **fixed** code

Generated Vitest tests are executed against both versions. A test that fails before and passes after is evidence the fix is real.



# Demo: a shopping-cart module with 5 seeded bugs.

**5**/<sub>5</sub>

seeded bugs caught

**5**

issues reported

**5**

issues fixed

**1 task**

end to end in Bob

Every number here is measured by our finalize script from a real Bob IDE run.  
Nothing is estimated.

# Every Bob run, replayable by anyone.

- Live agent pipeline with the real stage timings
- Issues with severity, type and line
- Side-by-side diff: original vs. fixed by Linty
- Executed tests: before → after, per issue
- Report built only from measured numbers, downloadable as JSON

# Every team that ships code reviews code.

## Who

Engineering teams, CI/CD owners and open-source maintainers who want faster, trustworthy reviews.

## Where it fits

Inside the IDE today via Bob; on every pull request next, via a GitHub Action.

## Model

Free for open source; per-seat for teams, with an audit trail of every AI change for regulated industries.



# Other tools suggest. Linty proves.

|                             | LINTERS    | AI CHAT ASSISTANTS | LINTY           |
|-----------------------------|------------|--------------------|-----------------|
| Finds logic bugs            | Rarely     | Yes                | Yes             |
| Applies the fix             | Style only | Suggests           | Yes, minimal    |
| Writes regression tests     | No         | On request         | Every fix       |
| Runs tests to prove the fix | No         | No                 | Before & after  |
| Scoped agent permissions    | n/a        | No                 | Per-agent files |

# Linty: review done, proof attached.

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